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7 December 1967
SUPERSEDING
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20 September 1954

MILITARY SPECIFICATION

TRAILER, MACHINE GUN MOUNT, M20

MOUNT, TRAILER MULTIPLE CAL. .50 MACHINE GUN, M55

1. SCOPE

1.1 Scope. This specification covers one type of two-wheel mount. The mount M45C is mounted on the one ton, two wheel machine gun mount, trailer M20. The combination is designated as the multiple caliber .50 machine gun trailer mount M55. The trailer M20 is a two wheeled vehicle designed to transport the mount M45C.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

FF-B-187	Bearing, Roller, Tapered
TT-E-529	Enamel, Alkyd, Semi-gloss

Military

MIL-P-514	Plates, Identification, Instruction, and Marking, Blank
MIL-M-13595	Mounts, Machine Gun; Multiple Cal. .50, M45, M45C, M45D, and M45F
MIL-V-45349	Vehicles: Combat and Special Purpose; Tactical, Self-Propelled and Towed, General Specification for

FSC 2330

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STANDARDS

Military

MIL-STD-193	Painting Procedures, Tactical Vehicles (Tracked and Wheeled)
MIL-STD-642	Identification Marking of Administrative, Combat, and Tactical Transport Vehicles
MIL-STD-1261	Welding Procedures for Constructional Steels

DRAWINGS

Army - WC

C5584934	Lamp, Tail, Right, Med. Tank M3
B6198559	Cable, Jumper
C7011684	Jack Assembly
D7015630	Trailer Assy, Mount M20
D7015634	Wheel and Bracket Assembly
D7015643	Trailer Mount, M55, Multiple Cal. .50 Mach. Gun
K7015697	Mount, Machine Gun, Mult. Cal. .50, M45C

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Standard Welding Symbols
Standard AWS A2.0

(Contractor may obtain copies of Industry Standards from the American Welding Society, 345 East 47th Street, New York, New York 10017.)

(Military Activities may obtain copies of Industry Standards from the Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

3. REQUIREMENTS

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3.1 Preproduction sample. As soon as practicable after the award of a contract or order and before proceeding with production in quantity, the contractor shall furnish one complete trailer, or trailer with mount, as applicable, for test to determine conformance to this specification. Approval of the preproduction sample shall not relieve the manufacturer of his responsibility to supply trailers, or trailer with mounts, as applicable, conforming to this specification.

3.2 Materials. Materials shall be as specified on the applicable drawings of Drawing D7015643. Materials not specifically covered shall have physical qualities adequate to conform to all the requirements of this specification.

3.3 Interchangeability. All components shall be constructed in accordance with the applicable drawings of Drawing D7015643. Parts shall be constructed so that any part may be replaced, fitted, and adjusted without modification.

3.4 Weight. Gross weight of the complete trailer with mount shall be approximately 3,000 pounds.

3.5 Painting. Parts and assemblies shall be prepared and painted as specified on the applicable drawing and in accordance with MIL-STD-193 and this specification.

3.6 Marking.

3.6.1 Registration numbers. Each trailer shall have registration numbers in block letters and figures, 3-inches high, stenciled in white semi-gloss enamel, conforming to TT-E-529, in approved locations. Registration numbers will be furnished by the procuring agency (see 6.2) and in accordance with MIL-STD-642.

3.6.2 Data plates. Name, data, caution, or other informative plates shall conform to the applicable drawings and shall be marked with the information as furnished by the procuring activity, and attached in the specified locations. No manufacturer's emblems or name plates shall be placed on the trailer. Identification plates for trailers shall be in accordance with MIL-P-514.

3.7 Gun mount. The gun mount M45C, Drawing K7015697, shall be installed on the trailer M20, by the contractor. Such installation shall cause no malfunctions of the mount and components and shall have complete accessibility to these units. Trailer and gun shall be subjected to the test requirements specified in 4.5.

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3.8 Chassis frame. The chassis frame of mount trailer M20, Drawing D7015630 shall be a box section, formed of steel plates as specified on the applicable drawings. The plates shall be so joined together by welding (see 3.18) as to prevent separation of members upon impact, or distortion of the frame unit when operating under military conditions. A suitable flanged mounting ring shall be secured on the top of the frame for installing the machine gun mount.

3.9 Drawbar. Each trailer shall be provided with a drawbar as indicated on Drawing D7015630, the construction of which shall provide for maneuverability by a crew of men or towing by a transport vehicle. Drawbar shall be of tubular construction and shall be securely welded to the chassis frame. A lunette, mounted on the drawbar shall be of forged or cast steel.

3.10 Wheel and axle mounting. Wheels shall be mounted on stub axles and tapered roller bearings in accordance with Drawing D7015634. Wheel and axle assemblies shall be attached to the frame in such a manner that they can be quickly and securely mounted, in preparation for towing and dismounted for firing. Mating parts of axle brackets and frame brackets shall assemble and disassemble smoothly and without interference. The means employed for securing the axles to the frame shall lock them securely.

3.10.1 Wheel bearings. Wheels shall be mounted on tapered roller bearings conforming to type 751 of FF-B-187. The bearings shall be of sufficient capacity to carry the load at the specified speeds (see 4.5.1). A positive means shall be provided for making minor bearing adjustments. All bearings shall be adequately protected by seals specified on the applicable drawings.

3.11 Tires and tubes. Tires shall be pneumatic, smooth-contour tread, size 22 x 7.25 x 11.50, six ply, aircraft type in accordance with the applicable drawing. Tires shall be free from cuts, cracks, or other evidence of damage. Inflation pressure shall be 50 pounds.

3.12 Jacks. Trailers shall be equipped with three manually operated jacks, as specified on Drawing C7011684, placing one at each rear corner of the frame and one on the drawbar. The jacks shall have sufficient travel to lift the trailer from the ground, and after removal of the wheels, lower the trailer to the ground for firing position. Jacks shall function smoothly and also lock securely in position for traveling.

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3.12.1 Jack handles. Three jack handles shall be provided in conformance with the applicable drawings. Jack handles shall fit the drawbar handles to form an extension thereof.

3.13 Wiring. Cable with plug, conforming to Drawing B6198559 shall be provided and connected to the rear lamp. A suitable arrangement shall be provided for protecting the plug and for carrying the cable slack.

3.14 Lamp. One combination black-out tail and stop lamp conforming to Drawing C5584934 shall be mounted on the left rear of trailer.

3.15 Reflectors. A reflector of the type, size, and color specified on the applicable drawing shall be installed on the rear access door and one each on the rear side frame members, as specified on the applicable drawing.

3.16 Tools. Tools shall be furnished by the contractor as specified by the procuring activity.

3.17 Lubrication. The chassis lubrication system shall be of the high-pressure type and shall be in accordance with the applicable drawings. Each trailer shall be thoroughly lubricated prior to delivery.

3.18 Welding. Welding shall be in conformance to the applicable drawings and in accordance with MIL-STD-1261. Welding symbols shall be interpreted, as specified in AWS A2.0.

3.19 Performance. The fully equipped trailer or trailer with mount shall be so constructed, that it will withstand being towed over relatively smooth terrain at speeds of 10 to 15 miles per hour for a distance of not less than 5 miles, and over all types of irregular terrain at speeds up to and including 5 miles per hour for a distance of not less than 10 miles.

3.20 Workmanship. Workmanship shall be in accordance with MIL-V-45349.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements

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as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample of a complete trailer or trailer with mount as specified in 3.1 shall meet the test requirements as specified in 4.5. Failure of the preproduction sample to pass the specified test shall be cause for the Government to reject the trailers, or trailers with mounts as applicable, until the cause of the rejection is corrected to the satisfaction of the procuring activity.

4.3 Sampling.

4.3.1 Lot size. Unless otherwise specified in the contract or order, a lot shall consist of 100 or less trailers or trailers with mounts, as applicable, submitted at one time by one manufacturer for acceptance.

4.3.2 Sample procedure. Unless otherwise specified in the contract or order, random samples of not less than 10 percent of each lot submitted shall be taken for acceptance.

4.4 Inspection. To determine conformance to this specification inspection shall be conducted in accordance with the applicable requirements of MIL-V-45349. Unless otherwise specified in the contract or order, inspection shall be at the place of manufacture. Prior to submission of the trailer or trailer with mount, as applicable for acceptance, the manufacturer shall maintain in-process inspection at appropriate stages of manufacture to determine conformance to this specification.

4.4.1 Wheel assemblies. The wheel and spindle assemblies shall be inspected for ease and rapidity of mounting and dismounting.

4.4.2 Tires. The tires shall be checked for proper inflation (see 3.11), with an approved gage and shall be inspected for cuts, cracks, or other injurious damage.

4.4.3 Tools. Tools shall be inspected and checked for completeness and proper storage.

4.4.4 Lamps. The combination blackout, tail and stoplight shall be inspected for proper location and proper functioning (see 3.14).

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4.5 Test procedures. Each trailer or trailer with mount, as applicable, submitted for tests shall be completely assembled, lubricated, equipped and loaded to the specified gross weight.

4.5.1 Road test. The first trailer produced under an order or contract, and not less than one percent of the trailers produced thereafter, shall be subjected to a road test. The trailer shall be towed by a suitable vehicle over rough terrain where the depth of ruts and holes do not exceed the ground clearance, at speeds up to and including 5 miles per hour, for a distance of not less than 10 miles. The trailer shall also be towed on good roads at speeds of 10 to 15 miles per hour, for a distance of not less than five miles. For this test, the trailer shall be fully equipped and loaded to "combat loaded" weight or equivalent, loading shall be provided by means of distributed weight. There shall be no damage to trailer or mount or components.

4.5.2 Proof firing. After completion of the road test specified in 4.5.1, the complete unit shall be subjected to a proof firing test as specified in MIL-M-13595. Operation of the mount shall be by means of slave batteries. The engine shall not be started.

4.5.3 Evaluation of tests. After each of the tests specified in 4.5.1 and 4.5.2, the trailer or trailer with mount, as applicable shall be thoroughly examined for evidence of defects of any kind.

4.5.4 Rejection. Failure of assemblies to pass either of the performance tests shall be cause for rejection of the subsequent lots until cause of failure is corrected.

4.5.5 Resubmission. The rejected lots shall be screened and all defects rectified before resubmission. The Government shall be notified of all action taken to prevent recurrance of the defects causing failure. Each resubmitted lot shall be clearly identified as such.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use. The trailers covered by this specification are intended to provide mobility for the M45C multiple gun mount. These trailers may be towed by army vehicle having a pintle-type trailer hitch and may be maneuvered by manpower.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Preproduction sample (3.1).
c. List of drawings (2.1).
d. Level of preservation, packaging, packing and marking (see 5.1).
e. Registration numbers (3.6.1).
f. Tools to be furnished (3.16).

Custodian:
Army - WC

Reviewer activity:
Army - WC

User activity:
Army - AT

Preparing activity:
Army - WC

Project No. 2330-A406

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